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Alerting effects of short-wavelength (blue) and long-wavelength (red) lights in the afternoon

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HIGHLIGHTS

- Light can be used to increase alertness in the afternoon, close to the post-lunch dip hours.
- Red light is a stronger alerting stimulus in the afternoon than blue light.
- Long-wavelength cones mediate the alerting effects of red light during the daytime.

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ABSTRACT

Light has an acute effect on neuroendocrine responses, performance, and alertness. Most studies to date have linked the alerting effects of light to its ability to suppress melatonin, which is maximally sensitive to short-wavelength light. Recent studies, however, have shown alerting effects of white or narrowband short-wavelength lights during daytime, when melatonin levels are low. While the use of light at night to promote alertness is well understood, it is important to develop an understanding of how light impacts alertness during the daytime, especially during the post-lunch hours. The aim of the current study was to investigate how 48-minute exposures to short-wavelength (blue) light (40 lux, 18.9 microWatts/cm^2 λ_{max} = 470 nanometers [nm]) or long-wavelength (red) light (40 lux, 18.9 microWatts/cm^2 λ_{max} = 630 nm) close to the post-lunch dip hours affect electroencephalogram measures in participants with regular sleep schedules. Power in the alpha, alpha theta, and theta ranges was significantly lower ($p < 0.05$) after participants were exposed to red light than after they remained in darkness. Exposure to blue light reduced alpha and alpha theta power compared to darkness, but these differences did not reach statistical significance ($p > 0.05$). The present results extend those performed during the nighttime, and demonstrate that light can be used to increase alertness in the afternoon, close to the post-lunch dip hours. These results also suggest that acute melatonin suppression is not needed to elicit an alerting effect in humans.

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1. Introduction

The endogenous circadian pacemaker, which is synchronized daily with the environment by the 24-hour light/dark patterns incident on the retina, sets the timing of circadian rhythms. In diurnal species, the circadian pacemaker promotes alertness during the day and sleepiness at night. The timing of the circadian system and the duration of time awake (sleep homeostasis) influence alertness and performance [1].

The duration of time awake reflects a homeostatic process, resulting in an increase in sleep pressure as the number of waking hours increases. Therefore, stable and high levels of alertness and consolidated sleep can be maintained when the phase relationship between the internal circadian timing system opposes the homeostatic process to promote daytime alertness and nighttime sleepiness. The circadian drive for daytime alertness, however, is not sufficiently strong to oppose the increased homeostatic sleep pressure during one specific time of day, which is referred to as post-lunch dip [2,3]. In fact, it has been shown that the timing of this dip in alertness varies between 14:00 and 16:00 (for a review, see [4]).

Human error is estimated to be a causal or contributing factor in 45–90% of accidents [5]. Acute or chronic sleep deprivation resulting in increased feelings of fatigue is one of the leading causes of human error and accidents in the workplace [6]. The daytime distribution of incidents and performance failures throughout the 24-hour day suggests a time of day effect with two peaks that can be seen in the early morning and mid-afternoon (post-lunch dip) hours [7].

Previous studies have shown that in addition to resetting the timing of the circadian pacemaker, light can have an acute effect on neuroendocrine responses, performance, and subjective and objective alertness. Most studies examined the effects of light on alertness at

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night, when the detrimental effects of fatigue can be easily observed. Compared to dim light or darkness, light at night (>2500 lx of white light) increased task performance, elevated core body temperature, and increased heart rate [8,9]. Light at night has also been shown to increase high frequency (beta range) and reduce low frequency (theta and alpha ranges) electroencephalographic (EEG) activity, as well as reduce slow-eye movements and subjective sleepiness, all measures that have been associated with decreasing sleepiness and increasing alertness at night [8–10].

The impact of nocturnal exposure to low levels (>5 lx) of short-wavelength (blue) light on subjective and objective alertness, performance, and biomarkers (melatonin and cortisol) has also been investigated [11–13]. Melatonin is a hormone produced at night and under conditions of darkness, and is used as a marker of the circadian system. It is now well accepted that the circadian system, as measured by acute melatonin suppression, is maximally sensitive to short-wavelength (blue) light; therefore, lower levels of short-wavelength (as low as 5 lx) light have been shown to have a similar impact on melatonin levels, objective and subjective alertness, and performance to higher levels (>2500 lx) of white light.

In most studies to date, the alerting effects of light have been linked to its ability to suppress melatonin [13]. In a recent study, however, Figueiro et al. [14] demonstrated that exposures to both short-wavelength (blue) and long-wavelength (red) lights in the middle of the night increased beta and reduced alpha power relative to preceding dark conditions, although only blue light significantly suppressed melatonin levels relative to darkness. Exposures to high (i.e., 40 lx at the cornea), but not low (i.e., 10 lx at the cornea), levels of red and of blue light significantly increased heart rate relative to darkness. Their findings suggest that the acute melatonin suppression is not needed for light to have an affect on measures of alertness at night. These results are consistent with studies showing alerting effects of light during the daytime, when melatonin levels are low [15,16].

While the use of light at night to promote alertness in the graveyard shift is well understood, it would be important to also develop lighting schemes that will promote alertness during the daytime, especially during the post-lunch hours. The aim of the current study was to extend the findings from Figueiro et al. [14] and investigate how exposure to short-wavelength blue light ($\lambda_{max} = 470$ nanometers [nm]) and long-wavelength red light ($\lambda_{max} = 630$ nm) during the middle of the afternoon, close to the post-lunch dip hours, affects objective measures of alertness and subjective measures of sleepiness in participants with regular sleep schedules. It was hypothesized that, if light can impact alertness via pathways other than melatonin suppression, as suggested by Figueiro et al. [14], both blue and red lights would decrease alpha, theta, and alpha theta power as well as decrease self reports of sleepiness relative to dim light.

2. Materials and methods

2.1. Participants

Thirteen healthy, paid volunteers, eight males (19–28 years, median = 20.5) and five females (18–25 years, median = 21), who met the following criteria were accepted and completed the study: (1) neither extreme late nor early type according to their responses on the Munich ChronoType Questionnaire [17], (2) no report of any physical or mental health problems (neither diagnosed with bipolar disorder nor depression) (3), no experience of shift work or travel to a different time zone within the 3 months prior to the experiment, and (4) no color blindness, as measured by the Ishihara for color blindness test. Participants were asked to refrain from alcohol and caffeine intake 12 h before the experimental session. Participants were also asked go to bed between 22:00 and 23:00 the night before the experiment and wake up no later than 7:30, and refrain from napping on the day of the experiment. Compliance was verified with sleep logs that participants were asked to keep during the experiment weeks. Participants were required to keep this sleep schedule during the entire experimental period to assure that measurements were performed close to the timing of their post-lunch dip, which typically occurs 16 to 18 h after bedtime, from 14:00 to 16:00 [4]. Upon arrival to the laboratory, participants signed an informed consent form approved by the Institutional Review Board at Rensselaer Polytechnic Institute.

2.2. Lighting conditions

In addition to dark (<0.01 at cornea), two experimental lighting conditions: long-wavelength "red" ($\lambda_{max} = 630$ nm) and short-wavelength "blue" ($\lambda_{max} = 470$ nm) were delivered to the corneas of each participant by arrays of light emitting diodes (LEDs) with a full width at half maximum of 25 nm placed in 60 × 60 × 60 cm light boxes. The LEDs (Lumileds, Luxeon I) were hidden from the direct view of the participants, and light boxes were painted white, so that a uniform and non-glaring distribution of light was achieved. Temperature inside the light boxes was constant throughout the experiment (range was between 68 and 70 °F). Before the experiment, each of the light boxes was calibrated using a Gigahertz-Optik X91 illuminance photometer to provide 40 lx of blue (40.2 μW/cm^2) and red (18.9 μW/cm^2) lights at the plane of the participant's cornea. It has already been established that the acute alerting effect of light has a broader spectral sensitivity than acute melatonin suppression, which is maximally sensitive to light peaking at 460 nm and insensitive to light peaking at 630 nm. Therefore, we equated the stimuli according to the orthodox photopic luminous efficiency function (lux), which has sensitivity at long-wavelengths. This does not imply, however, that the spectral sensitivity of the acute alerting effect of light is accurately characterized by this spectral weighting function.

2.3. Electroencephalogram (EEG)

The BioSemi ActiveTwo system (BioSemi, Amsterdam, Netherlands) with active electrodes was used for EEG recordings. Electrodes were placed on participant's scalps according to the International 10–20 system at Fz, Cz, Pz, and Oz. Two additional electrodes were attached to the right and left earlobes to serve as reference electrodes for those attached to the scalp. The BioSemi system replaces the ground electrodes, which are used in conventional systems, with the Common Mode Sense (CMS) active electrode and the Driven Right Leg (DRL) passive electrode, both of which are placed on the forehead of the participant. These two electrodes form a feedback loop, which drives the average potential of the subject (the Common Mode voltage) as close as possible to the analog-to-digital reference voltage in the analog-digital box. In order to monitor eye blinks, one electrode was placed directly below the right eye.

2.4. Subjective sleepiness (KSS)

To obtain information on subjective sleepiness, the Karolinska Sleepiness Scale (KSS) was rated four times (at the beginning, 12 min, 36 min, and at the end of the session). The KSS scale ranged from 1 to 9, where 1 = "very alert," 3 = "rather alert," 5 = "neither alert nor sleepy," 7 = "sleepy, but no difficulty remaining awake," and 9 = "very sleepy, fighting sleep, an effort to remain awake."

2.5. Procedure

Participants experienced three experimental sessions, separated by one week. Participants were asked to come to the Lighting Research Center laboratory at 14:00 to get instructions and be fitted with scalp electrodes for EEG recordings. All participants reported having eaten between 60 and 90 min prior to arriving at the lab.

Participants were kept seated under the dim light condition (<2 lx at eye level) during this preparation period. Sessions started at 14:30, and lasted for approximately 60 min. During the study, each participant was presented dark (D, <0.01 lx), red (RL, $\lambda_{max} = 630$ nm, 40 lx), and blue (BL, $\lambda_{max} = 470$ nm, 40 lx) lighting conditions. The presentation order of the three lighting conditions was counterbalanced across the participants. Under the D condition, participants were kept in front of the light boxes with the non-energized LEDs for 60 min. Under RL and BL conditions, red and blue lights inside the boxes were energized for 48 min, preceded by a 12-min dark (<0.01 lx) period, when lights were not energized. Two-and-a-half min of EEG activities were recorded seven times during the 60-min period (once when LEDs were not energized and six times under each lighting condition D, RL, and BL). Participants were asked to look inside the boxes at all times and keep their eyes open for the duration of the experiment. The experimenters assured that participants had their heads aligned with the aperture of the light box and that the eyes were open at all times during the light exposure. Participants were instructed to put their chins on the chinrest when they heard a short tone audio signal, which was presented via earphones 10 s before each of the EEG recordings. Long tone audio signals indicated the start and end of each EEG recording, and participants were instructed to sit still, stare at a printed X at the far end of the box, place their chin on chin rests, not move their heads and bodies, and avoid blinking as much as possible. In between each EEG recording, participants were instructed to remove their chins from the chinrest and relax for 5.5 min while keeping their heads aligned with the boxes to assure proper light exposures. Subjective sleepiness was evaluated using the 9-point KSS four times (at the beginning, 12 min, 36 min, and at the end of the session). Participants were informed when to fill out the forms that were located next to each light box by the three consecutive short auditory signals. Fig. 1 summarizes the experimental protocol and Fig. 2 shows the experimental apparatus, including the light boxes used to deliver light to the participants' corneas.

Fig. 2. Experimental apparatus. Participants were asked to look inside the boxes at all times and keep their eyes open for the duration of the experiment. During the EEG measurements, participants were asked to keep their chin on the chin rest to avoid head movements and look at an X printed inside the box, to avoid eye movements.

2.6. Data analyses

All subsequent EEG data analyses were performed with MATLAB, version 2010a. Values from the two reference channels were averaged together and subtracted from all other channels. A 0.3–40 Hz bandpass filter was used to remove signals outside of the relevant frequency band.

Data were then grouped into two-second epochs with a one second overlapping from each epoch (e.g., epoch 1 is 0–2 s, epoch 2 is 1–3 s, etc.). Any epochs with fluctuations exceeding ± 100 μV on the Fz, Pz, Cz, or Oz channels or 40 μV on the electrooculogram (EOG) channel were removed from the analysis. A 10% cosine window followed by a fast Fourier transform was then applied to each epoch. This process yielded spectral power distributions from 0.3 to 40 Hz in 0.5 Hz intervals. The power spectra from each epoch were then averaged together to yield average power spectra for each trial. These power spectra from each four electrode sites were grouped into the following bins: 4–8 Hz (theta), 5–9 Hz (alpha theta), and 8–12 Hz (alpha).

In each frequency range, EEG power at each interval was normalized to the power obtained during the initial dark adaptation trial. Subjective ratings of sleepiness (KSS scores) were also normalized to the initial dark adaptation trial. For the EEG measurements, a 3 (lighting conditions) × 4 (electrode sites) × 6 (trials) repeated-measures ANOVA was performed using the normalized power in each of the frequency ranges studied (alpha, alpha theta and theta). For the KSS, a 3 (lighting conditions) × 3 (time) ANOVA was performed. Two-tailed paired Student's *t*-tests (with Bonferroni corrections when needed) were used to further compare the main effects and interactions. All statistical analyses were performed using PASW Statistics 18 (Chicago, IL).

3. Results

3.1. EEG

A 3 (lighting conditions) × 4 (electrode sites) × 6 (trials) repeated-measures ANOVA revealed a significant main effect of lighting

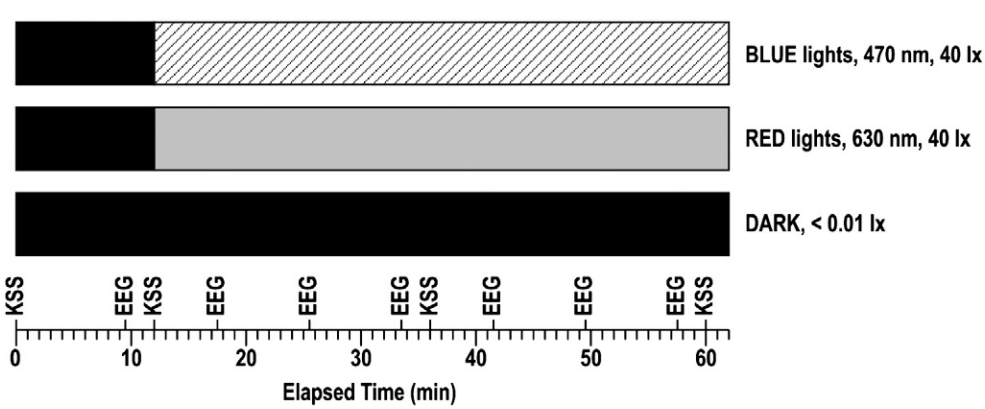

Fig. 1. Experimental design. The elapsed time from the start of the study is shown at the bottom of the graph. EEG data were recorded at 9.5th (T1), 17.5th (T2), 25.5th (T3), 33.5th (T4), 41.5th (T5), 49.5th (T6), and 57.5th (T7) min of the experiment and lasted for 2.5 min. KSS scores were obtained from subjects at the start of each experimental condition, and just after the 1st, 4th, and 7th EEG recordings.

Table 1
Pairwise comparisons for the main effect of lighting conditions.

		Pairs	t	df	Sig. (2-tailed)*
Main effect of lighting	Alpha	Dark–blue	1.254	12	0.234
		Dark–red	3.860	12	0.002*
		Blue–red	2.296	12	0.040
	Alpha theta	Dark–blue	0.407	12	0.691
		Dark–red	7.378	12	0.000*
		Blue–red	2.340	12	0.037
	Theta	Dark–blue	−0.158	12	0.877
		Dark–red	6.579	12	0.000*
		Blue–red	2.239	12	0.045

* Statistically significant. Bonferroni correction was applied ($p \leq 0.017$ are considered statistically significant).

conditions in the normalized alpha ($F_{2,24} = 5.6$; $p = 0.01$), alpha theta ($F_{2,24} = 6.1$; $p = 0.007$), and theta ($F_{2,24} = 4.6$; $p = 0.02$) ranges. Power in the alpha, alpha theta, and theta ranges was significantly lower after exposure to the RL condition than after exposure to the D condition. *Post-hoc* Student's *t*-tests did not reveal significant differences in power in the alpha, alpha theta, and theta ranges between BL and D conditions. After applying the Bonferroni correction, *post-hoc* Student's *t*-tests did not reveal significant differences in power in the frequency ranges studied between BL and RL conditions. Table 1 lists the p values for the main effect of lighting condition pairwise comparisons. Figs. 3a, 4a, and 5a show the significant main effects of lighting for alpha, alpha theta, and theta ranges.

A significant main effect of trials was also observed for the normalized alpha ($F_{5,60} = 5.6$; $p < 0.0001$), alpha theta ($F_{5,60} = 7.5$; $p < 0.0001$), and theta ($F_{5,60} = 4.4$; $p = 0.002$) ranges. In general, subjects became less alert over the course of the experiment. *Post-hoc* Student's *t*-tests were performed and revealed that power in the alpha, alpha theta, and theta ranges was significantly greater at trials 5, 6 and 7 than at trial 2. In addition, power in the theta range was significantly greater at trial 3 than at trial 2. Table 2 lists the p values for the main effect of trials pairwise comparisons.

None of the interactions between the variables were statistically significant ($p > 0.05$). Figs. 3b, 4b, and 5b show the normalized power at each trial for each frequency range studied. *Post-hoc* Student's *t*-tests were performed to compare normalized power for RL and D conditions and for BL and D conditions at each trial. Power in the alpha and alpha theta ranges was significantly lower after the RL than after the D condition at trials 2, 3, 4, 5, 6, and 7. Power in the theta range was significantly lower after the RL than after the D condition at trials 2, 3, 5, and 7. Table 3 lists the p values for the lighting conditions by trial interaction pairwise comparisons.

3.2. Subjective sleepiness (KSS)

Subjects reported feeling sleepier at the end of the experiment compared to the start of the experiment, although the ANOVA did not show a significant main effect of time ($F_{2,24} = 1.3$; $p > 0.1$). There was also no significant main effect of lighting conditions ($F_{2,24} = 1.51$; $p > 0.1$) and no significant interactions ($F_{4,48} = 0.527$, $p > 0.1$) between the variables. Subjective ratings were, however, in the expected direction; that is they were lower (subjects reported feeling less sleepy) after exposure to the RL condition than after exposure to the D condition (Fig. 6a and b). The mean ± standard error of the mean (S.E.M.) KSS normalized scores were 1.5 ± 0.1 after exposure to the D condition, 1.2 ± 0.1 after

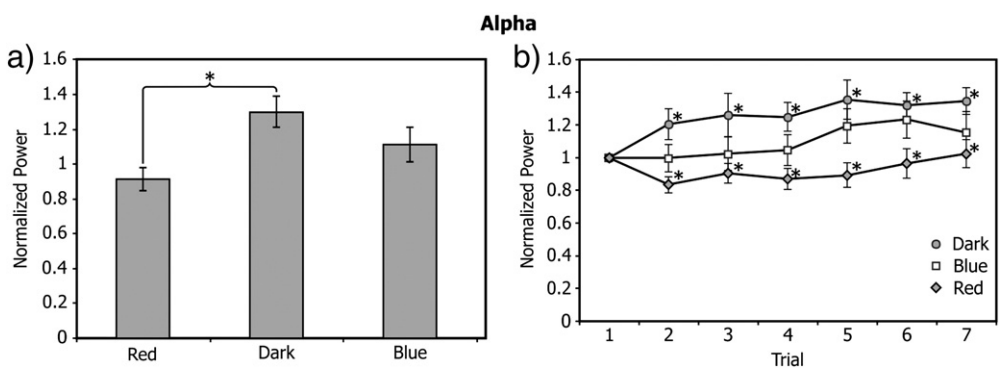

Fig. 3. a) Average ± standard error of the mean normalized alpha power (8–12 Hz) for RL, D, and BL conditions. Compared to remaining in darkness, alpha power was significantly reduced (*) after exposure to RL but not after exposure to BL; b) Average ± standard error of the mean of the normalized alpha power (8–12 Hz) for RL, D, and BL conditions at each trial (T1–T7). The power at trials 2 to 7 was normalized to the power at trial 1, which was collected in darkness. Compared to remaining in darkness, alpha power was significantly (*) reduced in the RL condition after approximately 5 min of light onset and remained significantly lower throughout the 48-minute light exposure period.

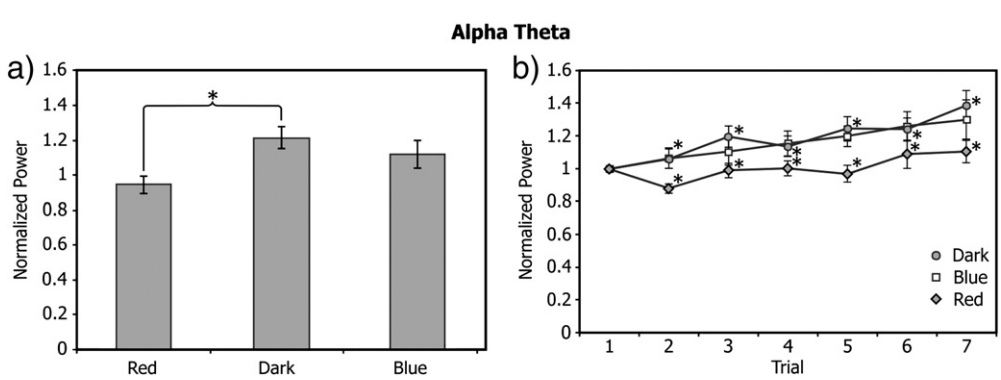

Fig. 4. a) Average ± standard error of the mean normalized alpha theta power (5–9 Hz) for RL, D, and BL conditions. Compared to remaining in darkness, alpha theta power was significantly reduced (*) after exposure to RL but not after exposure to BL; b) Average ± standard error of the mean of the normalized alpha theta power (5–9 Hz) for RL, D, and BL conditions at each trial (T1–T7). The power at trials 2 to 7 was normalized to the power at trial 1, which was collected in darkness. Compared to remaining in darkness, alpha theta power was significantly reduced (*) in the RL condition after approximately 5 min of light onset and remained significantly lower throughout the 48-minute light exposure period.

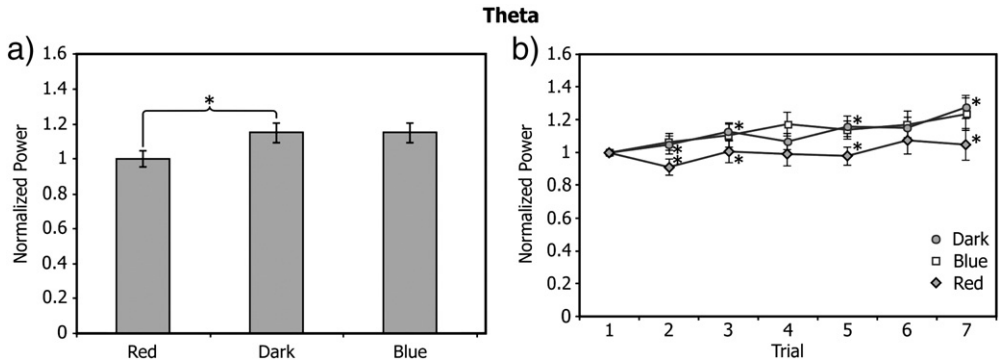

Fig. 5. a) Average ± standard error of the mean normalized theta power (4–8 Hz) for RL, D, and BL conditions. Compared to remaining in darkness, theta power was significantly reduced (*) after exposure to RL but not after exposure to BL; b) Average ± standard error of the mean of the normalized theta power (4–8 Hz) for RL, D, and BL conditions at each trial (T1–T7). The power at trials 2 to 7 was normalized to the power at trial 1, which was collected in darkness. Compared to remaining in darkness, there was an effect of RL on theta power after 5 min of light onset, but this effect was not sustained. Theta power was significantly lower after RL condition than after D condition at approximately 30 min after the onset of the light and then again at 42 min after light onset.

exposure to the RL condition and 1.5 ± 0.1 after exposure to BL condition (a higher normalized KSS score = more sleepiness).

4. Discussion

In the present study, we investigated how exposures to short-wavelength (blue) light (λ_{max} = 470 nm) and long-wavelength (red) light (λ_{max} = 630 nm) during the middle of the afternoon, close to the post-lunch dip hours, affect objective measures of alertness and subjective measures of sleepiness. Our results are the first to show that, compared to remaining in darkness, long-wavelength (red) light exposure in the middle of the afternoon significantly reduces power in the alpha, alpha theta, and theta ranges, suggesting that long-wavelength (red) light increases measures of alertness not only during the nighttime, as shown by Figueiro et al. [14], but also during the daytime hours.

Our results indicate that long-wavelength light is a more potent stimulus for modulating brain activities associated with daytime alertness than short-wavelength light. Compared to darkness, power in the alpha, alpha theta, and theta ranges after short-wavelength light exposure was reduced, but this difference did not reach statistical significance. It should be noted, however, that although power in the BL condition in all frequency ranges studied was not as low as those observed in the RL condition, these two conditions were not statistically different when the Bonferroni correction was applied to the *post-hoc* t tests. Therefore, the present results do not rule out the possibility that short-wavelength light is an alerting stimulus during the afternoon. Additional studies employing higher light levels of 470-nm light exposures or a larger sample size may lead to statistically significant changes in power in the alpha, alpha theta, and theta power ranges in the afternoon.

The present work extends from that by Figueiro et al. [14], who showed that both short- and long-wavelength light exposures in the middle of the night decreased alpha power and increased beta power compared to darkness. Here, we showed that the same long-wavelength light stimuli can also impact the objective measures of alertness in the middle of the afternoon. Vandewalle et al. [16] measured functional magnetic resonance imaging (fMRI) and showed

Table 2
Pairwise comparisons for the main effect of trials.

		Pairs	t	df	Sig. (2-tailed)*
Main effect of trial	Alpha	T2–T3	−1.066	12	0.307
		T2–T4	−1.460	12	0.170
		T2–T5	−3.192	12	0.008*
		T2–T6	−2.930	12	0.013*
		T2–T7	−3.991	12	0.002*
	Alpha theta	T2–T3	−2.738	12	0.018*
		T2–T4	−2.419	12	0.032*
		T2–T5	−3.184	12	0.008*
		T2–T6	−3.571	12	0.004*
		T2–T7	−4.078	12	0.002*
	Theta	T2–T3	−2.629	12	0.022*
		T2–T4	−2.094	12	0.058
		T2–T5	−2.909	12	0.013*
		T2–T6	−2.583	12	0.024*
		T2–T7	−3.690	12	0.003*

* Statistical significant at $p \leq 0.05$.

Table 3
Pairwise (*post-hoc*) comparisons for the lighting conditions × trials interaction.

		Pairs	t	df	Sig. (2-tailed)*
Lighting conditions × trial interaction	Alpha	DT2–BT2	1.523	12	0.154
		DT2–RT2	3.778	12	0.003*
		DT3–BT3	1.543	12	0.149
		DT3–RT3	2.703	12	0.019*
		DT4–BT4	1.428	12	0.179
		DT4–RT4	3.906	12	0.002*
		DT5–BT5	0.898	12	0.387
		DT5–RT5	3.450	12	0.005*
		DT6–BT6	0.553	12	0.590
		DT6–RT6	3.441	12	0.005*
		DT7–BT7	1.078	12	0.302
		DT7–RT7	2.737	12	0.018*
	Alpha theta	DT2–BT2	−0.037	12	0.971
		DT2–RT2	3.561	12	0.004*
		DT3–BT3	1.021	12	0.327
		DT3–RT3	3.113	12	0.009*
		DT4–BT4	−0.193	12	0.850
		DT4–RT4	2.338	12	0.038*
		DT5–BT5	0.509	12	0.620
		DT5–RT5	4.045	12	0.002*
		DT6–BT6	−0.212	12	0.836
		DT6–RT6	2.726	12	0.018*
		DT7–BT7	0.679	12	0.510
		DT7–RT7	4.766	12	0.000*
	Theta	DT2–BT2	−0.216	12	0.833
		DT2–RT2	3.195	12	0.008*
		DT3–BT3	0.281	12	0.784
		DT3–RT3	2.258	12	0.043*
		DT4–BT4	−1.320	12	0.211
		DT4–RT4	1.507	12	0.158
		DT5–BT5	0.300	12	0.769
		DT5–RT5	3.296	12	0.006*
		DT6–BT6	−0.227	12	0.824
		DT6–RT6	1.645	12	0.126
		DT7–BT7	0.399	12	0.697
		DT7–RT7	4.056	12	0.002*

* Statistical significant at $p \leq 0.05$.

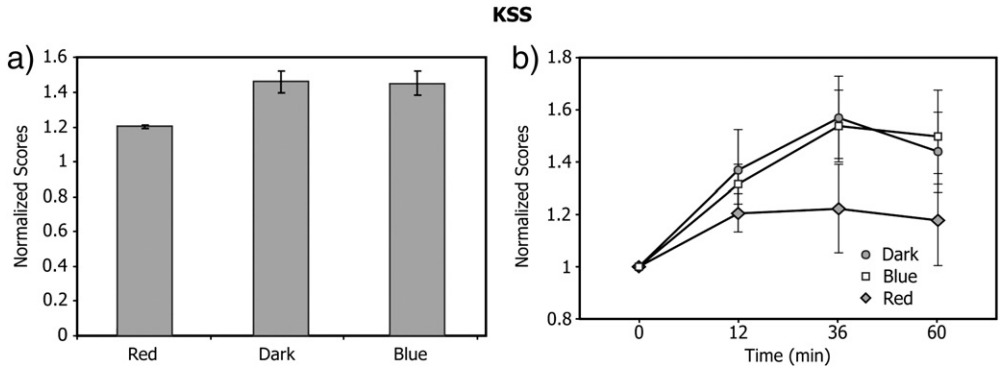

Fig. 6. a) Average ± standard error of the mean normalized KSS scores for RL, D, and BL conditions; b) Average ± standard error of the mean of the normalized KSS scores for RL, D, and BL conditions at each time. Although in the same direction as the EEG measures, the KSS scores after light exposures were not statistically different from the scores in darkness.

greater alerting effects after exposure to a 480-nm light than to longer (530-nm) or shorter (430-nm) light. These response differences may have been because EEG measurements are not as sensitive to brain activity changes as fMRI used in their study or because their light exposures occurred 5 h after habitual wake times while it occurred after 7 h in the present study; therefore, sleep homeostasis was greater in the present study. Moreover, they did not investigate the impact of longer wavelength (630-nm) light, such as the one employed in the present study.

Subjective sleepiness (i.e., KSS scores) reported after RL condition showed the same trend as the EEG measures, but were not statistically different than darkness. Rather than reflecting participants' self reports of sleepiness, the KSS scores may have been reflecting the boredom that they informally reported feeling after staring inside a box for nearly 1 h. Another explanation for these results may be that the correlation between subjective scores of sleepiness and objective measures of alertness is not as easily seen when subjects are not sleep deprived. Studies by Akerstedt and Gillberg [18] and Waage et al. [19] suggest that KSS is only associated with objective measures of sleepiness when sleep pressure is very high. In the present study, the highest mean ± S.E.M. reported KSS score was during the dark condition and it was 6.6 ± 0.4 (6 = some signs of sleepiness, 7 = sleepy, no effort to stay awake). Future studies should investigate whether the correlation between KSS scores and objective measures of alertness is greater when subjects are sleep deprived.

The present results together with previous work indicate that light-induced melatonin suppression is not needed for light to have an impact on objective measures of alertness. One study showed that a 5-h exposure to daytime light (1000 lx at the cornea) reduced subjective sleepiness, decreased slow eye movements, and improved psychomotor vigilance test performance compared to dim light, but bright light did not have an impact on daytime melatonin levels [15]. In another study [16], short exposure (21 min) to a very bright light (7000 lx) during the day prevented sleepiness that is associated with darkness and enhanced thalamic activity that may be modulating cortical responses to a cognitive task.

Further, while electroretinogram was not performed in the present study, it can be indirectly deduced from the present results that, while the intrinsically photosensitive retinal ganglion cells (ipRGCs) may participate in the light-induced alerting effects of light during the daytime, they are not the only photoreceptors directly mediating these alerting effects because the ipRGCs are maximally sensitive to short wavelengths (approximately 480 nm) [16] and very insensitive to long-wavelength light, such as the 630-nm light used in the present study.

From the present results, it is not possible to determine the underlying mechanisms contributing to light-induced changes in alertness because the optical radiation incident on the retina has multiple effects on brain activity through parallel neural pathways. For example, the decrease in theta activity by the long-wavelength light may reflect an effect of this kind of light on the homeostatic sleep system during the daytime. It has been demonstrated that theta activity increases with elapsed time awake [20], suggesting that the homeostatic component of sleep contributes more to the daily variation of theta activity than the circadian component. Another possibility is that this arousing effect of light is simply a psychological result of processing by neural mechanisms of the spectral irradiance distribution incident on the retina. Some studies suggest that colors seen as "warm" (red, orange, yellow) evoke feelings of arousal while colors seen as "cool" (violet, blue, green) are associated with calming feelings. It has been suggested that the color red increases human receptiveness to external stimuli and increases excitation, therefore affecting a person's emotional and motor responses [21,22]. Further studies designed specifically to test this hypothesis need to be performed.

5. Conclusions

The present results extend from those performed during the nighttime, and demonstrate that narrowband long-wavelength (red) light can be used to increase alertness in the middle of the afternoon, close to the post-lunch dip hours. Unlike what has been shown during the nighttime, short-wavelength (blue) light exposure in the afternoon had a smaller, albeit not statistically significantly different, effect on alertness than long-wavelength (red) light. Since the effect of short-wavelength light at night is well documented, future studies should be designed to further confirm a possible change in spectral sensitivity in light-induced alertness in humans over the course of 24 h.

Competing interests

The authors declare they have no competing interests.

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